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Projects

Stateful Conversational AI | LangChain, LangGraph, LangSmith, Streamlit, SQLite — Sep' 26 – Ongoing

- Built an LLM-powered conversational chatbot using **LangChain and LangGraph**, implementing state-based message handling and structured conversation workflows.
- Implemented **persistent conversation checkpoints using SQLite**, enabling users to create, switch between, and resume multiple conversation threads.
- Developed **real-time streaming responses** using LangGraph's streaming interface and integrated them into an interactive Streamlit chat interface.
- Integrated **LangSmith observability** with project-level tracing and run metadata to monitor, debug, and analyze chatbot executions.
- **GitHub repository:** Stateful Chatbot

VisionForge-AI | PyTorch, CNN, Transfer Learning — Sep' 26 – Ongoing

- Developing an end-to-end computer vision system to automatically identify and classify objects from images using deep learning and computer vision.
- Built the initial **data preprocessing, augmentation, training, validation, and inference pipeline** using PyTorch and established a baseline classification model.
- Evaluated model performance through **training/validation metrics and prediction-confidence analysis**, achieving high classification confidence on the current dataset.
- Currently scaling the system with a **larger and more diverse dataset**, with planned improvements including advanced architecture, robust evaluation, error analysis, and model optimization.
- **GitHub repository:** VisionForge-AI

Chat Intent Discovery using NLP Embeddings | Scikit-learn, NLTK, Gensim, Transformers — May' 26

- Designed an unsupervised NLP pipeline for automatic chat intent discovery by preprocessing customer-support conversations and grouping semantically similar messages without labeled data.
- Integrated multiple text representation techniques including TF-IDF, Word2Vec, GloVe, and Sentence-BERT embeddings to compare semantic understanding and clustering performance across traditional and transformer-based NLP models.
- Applied clustering algorithms such as K-Means, DBSCAN, and HDBSCAN, and evaluated cluster quality using Silhouette Score and Davies-Bouldin Score for effective intent separation.
- Visualized high-dimensional semantic embeddings using PCA and t-SNE to analyze intent separation, semantic similarity, and hidden conversational patterns in large-scale chat datasets.
- **GitHub repository:** Chat Intent Detection

Training

Data Structures and Algorithms + System Design — Jun' 26 – Jul' 26

- Completed structured training in **Data Structures & Algorithms**, focusing on algorithmic problem-solving, complexity analysis, optimization, and efficient coding techniques.
- Strengthened proficiency in **arrays, linked lists, stacks, queues, trees, graphs, recursion, dynamic programming, and greedy algorithms** through extensive practice.
- Developed optimized solutions to algorithmic problems with an emphasis on time and space complexity, efficient data-structure selection, and clean coding practices.
- Studied **System Design fundamentals**, including modular architecture, scalability, maintainability, and component-level design of software systems.
- Practiced and solved 1,000+ DSA problems on platforms including LeetCode and GeeksforGeeks, improving problem-solving accuracy, efficiency, and code optimization skills.

Certificates

Generative AI and LLMs: Architecture and Data Preparation — May' 26

Coursera — Certificate link

Improving Deep Neural Networks — May' 26

Coursera — Certificate link

Structuring Machine Learning Projects — May' 26

Coursera — Certificate link

Neural Networks and Deep Learning — Apr' 26

Coursera — Certificate link

Skills

Languages:	C++, Python, SQL
AI/ML:	Machine Learning, Deep Learning, NLP, Computer Vision, Transformers, LLMs, LLM Fine-Tuning
Generative AI:	LangChain, LangGraph, LangSmith, RAG, MCP, Prompt Engineering, Hugging Face
Libraries/Frameworks:	PyTorch, TensorFlow, Keras, Scikit-learn, NLTK, Gensim, NumPy, Pandas, Matplotlib
Databases/Tools:	MySQL, SQLite, Git, Linux, AWS, n8n
Soft Skills:	Analytical Thinking, Problem-Solving, Team Collaboration, Time Management

Education

Lovely Professional University **2023 – 2027**
Bachelor of Technology - Computer Science and Engineering — CGPA: 8.89 *Phagwara, Punjab*

Army Public School Dhaula Kuan **2021 – 2022**
12th — Percentage: 79% *Dhaula Kuan, Delhi*

Army Public School Bareilly Cantt **2019 – 2020**
10th — Percentage: 91.6% *Bareilly, Uttar Pradesh*